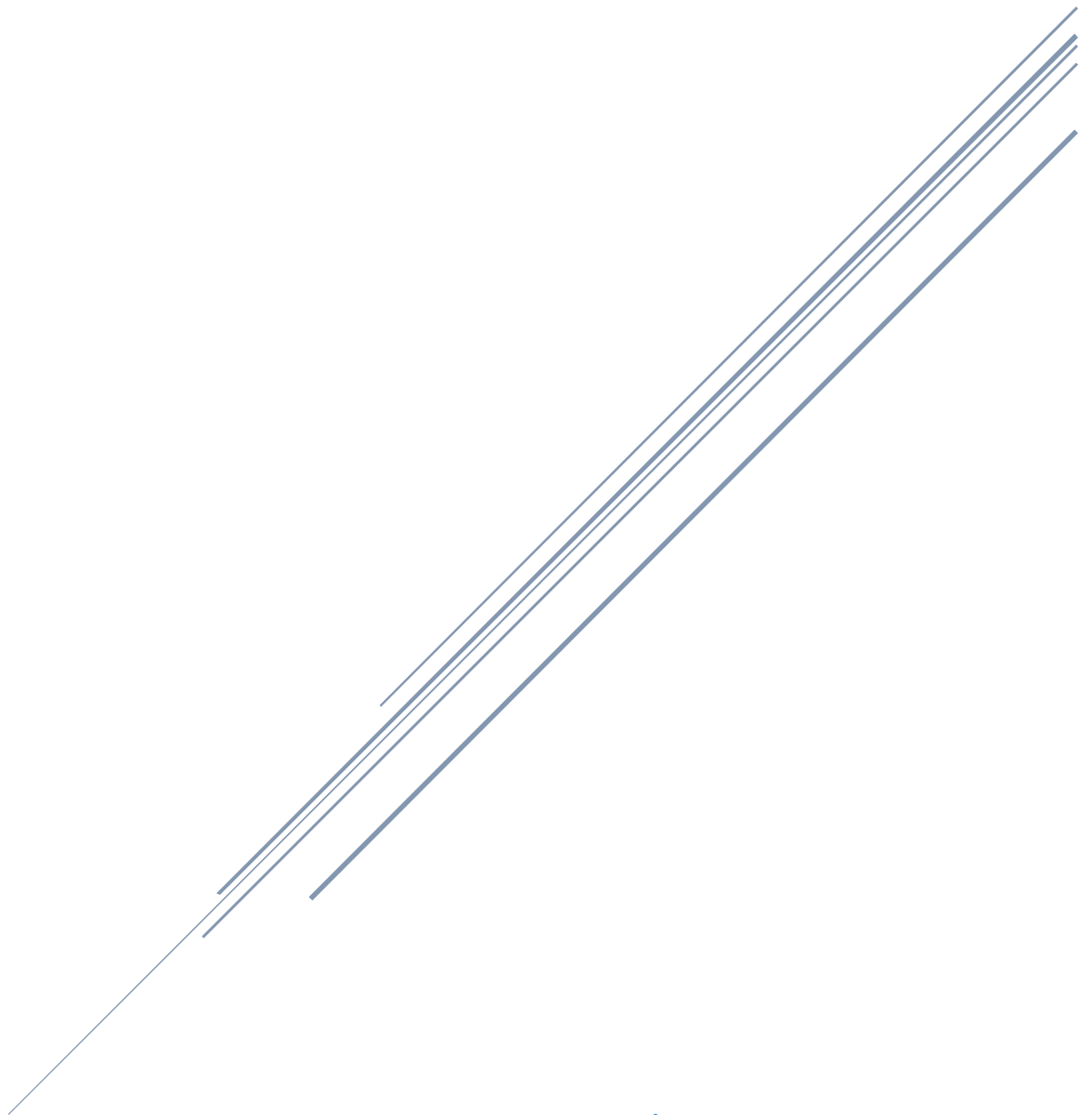


CODE AND SUMMARY OF CLASS 12

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Saturday (2 – 5)



Teachers:

Muhammad Bilal And Ali Aftab Sheikh

Code And Summary Of Class 12 – Saturday (2 – 5) | Quarter 3

Introduction

This document provides a detailed summary of Class 12 of Quarter 3, held on Saturday from 2 to 5 PM, led by Sir Ali Aftab Sheikh and Sir Muhammad Bilal. As the final class before the Eid break, the session introduced the upcoming Quarter 4 syllabus, focusing on AI agents, while continuing Quarter 3's Python curriculum. The class covered **AI agents, workflows, and Large Language Models (LLMs)**, and included a practical coding demonstration using the Open AI Agent SDK to create a simple agent. The session combined theoretical discussions, coding exercises, and motivational guidance to prepare students for the upcoming challenges of Quarter 4.

The class addressed the following key objectives:

- Introducing the Quarter 4 syllabus, focusing on AI agents, while completing Quarter 3's Python curriculum.
- Explaining AI agents, workflows, and LLMs, with a focus on practical application development.
- Preparing students for Quarter 4's self-directed learning and industry-relevant skills.
- Encouraging resilience, practical coding, and a startup mindset for market opportunities.

Students were informed that Quarter 3 is ongoing, with an exam and viva scheduled to transition to Quarter 4. Class timings may shift to 3–6:30 PM after Eid due to heat concerns. Quarter 4 will focus on AI agents and cloud computing, requiring students to adopt a self-directed learning approach.

Topics Covered

1. Class Updates and Quarter 3 Progress

- **Quarter 3 Status:** Class 12 continues Quarter 3's Python curriculum, with an upcoming exam and potential viva to prepare for the transition to Quarter 4. Students were encouraged to stay focused and prepare for advanced topics like AI agents.
- **Quarter 4 Logistics:** Due to heat concerns in June and July, class timings may shift from 2–5 PM to 3–6:30 PM after Eid (June 7). No classes will be held on Eid, resuming on June 14. Students can choose new slots post-Eid based on availability.
- **Motivational Advice:** Sir Ali emphasized resilience and consistent effort, urging students to explore repositories like the [Panaversity](#) repository and prepare for industry relevant skills. Students were encouraged to adopt a startup mindset, implement learned concepts, and target market opportunities.

2. Introduction to Quarter 4 Syllabus:

- **Syllabus Overview:** Quarter 4 will focus on AI agents, with the [Panaversity](#) repository as the primary resource. The course, titled “Panaversity Certified Agentic and Robotic AI Engineering,” shifts from Web 3 and Metaverse to AI and robotics. Students were advised to review the syllabus outline in the [Panaversity](#) repository.
- **Market Opportunities:** Completing Quarter 4 can lead to entry-level salaries of \$60,000–\$90,000 annually in international markets (e.g., USA, Canada). Students were encouraged to develop communication skills alongside technical skills to succeed in the industry.

3. AI Agents and Large Language Models (LLMs)

3.1 Overview of AI and LLMs

- **AI Fields:** AI was divided into two fields: researchers (who create models using machine learning, deep learning, and math) and application

developers (who use existing models to build applications). The class focuses on application development, using models like ChatGPT, Gemini, and LLMs.

- **LLM Functionality:** LLMs predict the next word based on input, converting text into numerical tokens for processing. They rely on pre-trained data and cannot make autonomous decisions without external tools.
- **Challenges of Model Creation:** Creating an LLM requires significant data, computing power (GPUs/TPUs), and investment, making it impractical for students. Instead, application developers use existing models to build solutions.

3.2 Workflows vs. Agents

- **Workflows:** Defined as a series of predefined steps (e.g., entering a mall, shopping, paying, exiting). Workflows are linear and do not involve decision-making.
- **Agents:** Use if-else conditions to make autonomous decisions, forming a decision tree. Agents run in a loop (while loop) until the desired output is achieved, breaking the loop upon completion.
- **Real-World Analogy:** Workflows are like following a fixed path in a mall, while agents decide paths based on conditions (e.g., if a store is open, shop there; else, go elsewhere).

3.3 Agent Components

- **Agent Loop:** Agents run continuously using a while loop with if-else conditions, processing input until the desired output is achieved. A token limit can be set to control resource usage.
- **Hand-Off:** Questions are handed off to specialized agents (e.g., a math agent for math queries) for better expertise, breaking the loop to delegate tasks.
- **Guards:** Act as filters to ensure agents only respond to relevant queries, rejecting unrelated inputs (e.g., a study agent rejecting non-study questions).

- **Tracing and Observability:** Logs agent actions (e.g., errors, tool calls) for debugging, stored in dashboards like Open AI's or separately for other models.

4. Practical Coding: Creating a Simple Agent

- **Setup:** Students used UV to create a project named “Hello Agent” and set up a virtual environment. The Open AI Agent SDK was installed to create a simple agent.
- **Coding Example:**

```
from dotenv import load_dotenv
from openai_agents import Agent, Runner

load_dotenv()

agent = Agent(name="Assistant", instructions="You are a
    helpful assistant.")
result = Runner.run_sync(agent, "What is PIAIC?")
print(result)
# Output: The Presidential Initiative for Artificial
    Intelligence and Cloud Computing is a program launched in
    Pakistan aimed at promoting education and skill
    development in key areas.
```

- **Explanation:** The code loads environment variables (Open AI API key), creates an Agent instance with a name and instructions, and uses the Runner class's run_sync method to process the input query (“What is PIAIC?”). The agent queries the LLM and returns the output. A test with “What is GIAIC?” returned no recognized meaning, demonstrating LLM limitations.
- **Dependencies:** The code uses the `openai_agents` and `python-dotenv` packages, installed via `uv add openai_agents` and `uv add python-dotenv`.
- **Agent Response And LLM Behavior:** The agent responded correctly to queries about PIAIC, but failed to answer questions about GIAIC, as it is not present in the LLM's training data, highlighting knowledge limitations.

5. Key Insights and Takeaways

- **AI Agent Development:** Understanding agents and their components (loop, hand-off, guards, tracing) is crucial for Quarter 4's application development focus.
 - **Self-Learning:** Quarter 4 will require independent learning using resources like the [Panaversity](#) repository. Students must explore and practice to master AI concepts.
 - **Industry Relevance:** AI agent skills are in high demand, offering significant career opportunities in international markets if combined with strong communication skills.
 - **Resilience:** Students were urged to overcome challenges (e.g., heat, complex concepts) and maintain consistent practice to build portfolios.
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6. 📖 Class Resources and Assignments

- **Today's Class Code**
The code from today's session is available here:
[Class Code Repository](#)
- **Assignment:** Smart Student Agent Assistant

What It Does

Build a simple Agent using **Python + OpenAI Agent SDK** that can:

- Answer academic questions
- Provide study tips
- Summarize small text passages

💡 *Keep it simple! This is your hands-on introduction to the OpenAI Agent SDK.*

- 📅 **Deadline:** Before the next class

- [📌 Submission](#): Students were urged to overcome challenges (e.g., heat, complex concepts) and maintain consistent practice to build portfolios.
 - Upload your assignment with screenshots or a demo video on **LinkedIn**
 - Submit the link via this [Assignment Submission Form](#)
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Important Reminder

- The next class resumes after Eid, continuing Quarter 3. Attendance is mandatory, and incomplete assignments may affect roles in Quarter 4.
 - Homework includes running the agent code with Gemini, studying SWARM, Open Router, and LLM, and creating an agent to demonstrate practical skills.
 - Stay active on Discord for class announcements and review the [Panaversity](#) repository for the syllabus.
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Final Words

Your responsibility is to learn actively, participate confidently, and practice consistently. Quarter 4 will demand greater effort with advanced topics like AI and cloud computing. Embrace challenges, complete assignments, and build your portfolio to seize IT industry opportunities. Stay focused, and may Allah guide you.

Allah Hafiz – See you in the Next Class!

Repo Link: [GIAIC-Sat-Afternoon](#)

This repository contains all class codes, assignments, and summaries of Quarter 3. Review each class's code and summary to prepare for upcoming classes.

Thank You for Reading!

Hope you understood Class 12 well.

"Education is not the learning of facts, but the training of the mind to think." – *Albert Einstein*